

PROF. DR. ATHER ALI KHAN

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QUALIFICATIONS: Ph.D "MARINE GEOLOGY" (1986-1989)
University of Edinburgh, U.K.

M.Sc " MARINE GEOLOGY". (1980)
First class 3rd Position
University of Karachi, Pakistan.

B.Sc (Hons) " GEOLOGY"(1979).
First class 3rd Position
University of Karachi, Pakistan

FIELDS OF SPECIALIZATION

Marine and Coastal Geology, Marine Geochemistry., Sea level changes and sedimentation. Climate changes and coastal zones, Quaternary studies of marine sediments. Environmental Geochemistry, Paleoclimate and Paleo-oceanography.

RESEARCH INTERESTS

Paleoceanography and Paleoclimatology, Monsoon variations and sedimentary records. Sea-level changes and impacts. Coastal and Marine Sedimentation.

AWARDS AND FELLOWSHIPS

1. Research Fellowship (December, 1998-May, 1999) awarded by Federal Ministry of Education and Research (BMBF, Bonn, Germany), to evaluate SONNE-130 cruise data and Geological samples from Arabian Sea at Federal Institute For Geosciences and Natural Resources (BGR), Hanover, Germany.

2. Research Fellowship (January-June, 1994) awarded by Geosciences Foundation, (GOA) of The Netherlands Organization for Scientific Research (NWO), The Hague, to work on Arabian Sea sediment cores collected in Netherlands Indian Ocean Program (NIOP-92).
3. British Government Technical Cooperation Training Award (1987-1989) for Postgraduate studies at Grant Institute of Geology and Geophysics, University of Edinburgh, UK.
4. British Government Technical Cooperation Training Award (1985-1986) for studies of shallow seismic data of North Arabian Sea at Imperial College of Science and Technology, University of London, U.K.

PROFESSIONAL EXPERIENCE

Professor Department of Geology, University of Karachi, Pakistan.
Approved HEC Supervisor

Member, Editorial Board. Journal of Basic and Applied sciences. University of Karachi.

Member Academic Committee Institute of Marine Science. University of Karachi

Visiting Faculty Member Institute of Space and Planetary Astrophysics. University of Karachi
(2004-2008)

Consultant (2005) Academic consultant to Pakistan Marine Academy. Karachi.

Course Organizer (2002-2008) Short course on Coastal Zone Management(CZM) for Managers
Policy makers and Researchers. In collaboration with the Centre of Excellence in Marine Biology, Karachi University and Scottish association for Marine Sciences, U.K.

Senior Research Officer National Institute of Oceanography, Pakistan.
(1990-June,2002)

Post-Doctoral Research Federal Institute For Geosciences and Natural Resources (BGR),
(1998-1999) Hanover, Germany

- Evaluation of SONNE-130 cruise data and Geological samples from Arabian Sea.
- Geochemistry and Petrography of Murray Ridge, Northeastern Arabian Sea
- Varve and Cyclostratigraphy, Oxygen Isotope and Chemical Composition of Holocene Sediments (Upper Makran slope, Pakistan)

- Post-Doctoral Research (1994)** University of Utrecht, The Netherlands.
- The Geological studies of the Arabian Sea.
- Instructor (1994)** IOC-IOMAC / advanced training course on Marine Geology and Geophysics off Pakistan.
- Co-Chief Scientist (1993)** Onboard Research Vessel SONNE in Pak-German Research cruise SO 90 (September –1993). The "PAKOMIN" Influence of the Oxygen Minimum Zone on the sedimentation at the upper continental slope off Pakistan. Funded by Federal Ministry of Research and Technology (BMFT, BONN) organized by Federal institute for Geosciences and Natural Resources (BGR) Hanover, Germany.
- Incharge Planning and Development (1993)** National Institute of Oceanography, Pakistan.
- Incharge Administration (1992)** National Institute Of Oceanography, Pakistan
- Co-Chief Scientist (1992)** The "NIOP" Netherlands Indian Ocean Program. Leg D2, The Geological studies of the Arabian Sea Project. "Late Quaternary Productivity and the Dynamics of the Oxygen Minimum Zone. National Institute of Oceanography. Pakistan
- Research Officer (1982-1990)**
- Research Scholar of (1986-1989)** Grant Institute of Geology and Geophysics, Edinburgh. University Edinburgh, U.K
- The Climatic reconstruction of the last 250,000 years of Arabian Sea. The effects of monsoon upwelling on the chemistry of sediments. Biogenic and Lithogenic fluxes, variations of N.W. Winds and Monsoon winds.
- Research Associate Technology, (1986-1987)** Department of Geology, Imperial College of Science and University of London, U.K.
- Acoustic Facies and Sedimentation in the Oman Basin and adjoining areas of Arabian Sea.

INTERNATIONAL RESEARCH PROJECTS

1. Nature and Timing of Sediment Flux Rates from the Indus River to the Arabian Sea. Collaborative Research Project with Department of Geology and Geophysics. Woods Hole Oceanographic Institute. Massachusetts. USA.
2. Makran II: The Makran Accretionary wedge off Pakistan. Tectonic Evolution and fluid Migration. Organized by Federal institute for Geosciences and Natural Resources (BGR) Hanover, Germany.

3. The "PAKOMIN" Influence of the Oxygen Minimum zone on the sedimentation at the upper continental slope off Pakistan. Funded by Federal Ministry of Research and Technology (BMFT, BONN) organized by Federal institute for Geosciences and Natural Resources (BGR) Hanover, Germany.
4. The "NIOP" Netherlands Indian Ocean Program. Leg D2, The Geological studies of the Arabian Sea Project "Late Quaternary Productivity and the Dynamics of the Oxygen Minimum Zone.
5. Geophysical and Geological Studies of Oman Basin, Owen Fracture Zone and Murray Ridge, Arabian Sea. Bullard Laboratories, University Of Cambridge, U.K.

PROJECTS AND TECHNICAL REPORTS

1. (2011) Supervised MAS project "Comparative mineralogical study of Post flood (2010) Indus river and Arabian Sea sediments.
(2008) Supervised a Ph.D research thesis Title " A Study of Dynamics of Upper High Salinity Water Mass in the Arabian Sea."
3. (2002-2008) Supervised M.Sc students reports of Marine Environments and Geological studies of coastal zones of Karachi.
4. (2003) Beach Sediments Physical Properties and Residual Oil Impacts (Tasman spirit Oil Spill 13th August-2003), Karachi.
5. (2001) Feasibility Study to Restore and Develop Bundal and Kiprianwala Islands. Pakistan Defense officers Housing Authority, Karachi.
6. (2000) Impact Assessment Study on Coastal Zones, Sea level Rise and Monsoon Variations. Pakistan National Communication.
7. (1999) Post cruise SO 130 Data Evaluation report "High resolution Palaeoclimatic studies (5000 years B.P to present) from varved sediments (Upper continental slope off the Makran, Pakistan). Submitted to BGR, Hanover, Germany.
8. (1998) SONNE Cruise SO 130- Makran II-Cruise Report.3rd April-3rd May 1998. Archaic Nr.117368. BGR Hanover, July1998. Germany.
9. (1998) NIO-BGR joint Field Report of Chandragup mud volcanoes and Hingol River section.
10. (1996) Hydraulic and hydrodynamic study for the reclamation and development of Bundal Islands. NIO Technical Report
11. (1995) Hydrographical and Environmental study, 206 MW steam turbine power plant, Abdullah Goth, Gadani, Baluchistan. M/S Star energy ventures (Pakistan) Limited.
12. (1995) Hydrographic survey for 134 MW Barge Mounted power station, Korangi creek, Karachi. NIO Technical Report (NIO TR 3/95.
13. (1994) Report; Post cruise work of Research vessel "Tyro" at Department of Earth Sciences, University of Utrecht, and The Netherlands.
14. **(1993) Shipboard Report, Research cruise "PAKOMIN" SO-90 with R.V Sonne. Influence of the Oxygen Minimum Zone on the sedimentation at the upper continental slope off Pakistan. Funded by Federal Ministry of Research and Technology (BMFT, BONN) organized by Federal institute of Geology and natural resources (BGR) Hanover, Germany.**

15. (1993) TA 1245(PAK) Marine Outfall Study, Karachi. (Geological Studies). Asian Development Bank funded project for Karachi water and Sewerage Board.
16. (1992) Shipboard Report, The Netherlands Indian Ocean Program (NIOP), Leg D2 "The Geological studies of the Arabian Sea".
17. (1992) Feasibility study for 350 MW FEPCO oil fired power station at Khalifa point. Hydrographic survey and Hydraulic investigations.
18. (1989) Geochemistry and Palaeoclimatic Changes in Sediments, North Arabian Sea.
19. (1986) Acoustic Facies and sedimentation in the Gulf of Oman, Arabian Sea.
20. (1984) Distribution of surface sediments and their carbonate content off Indus Delta continental shelf. NIO Tech Report.
21. (1982) Textural study of the Clifton beach sand, Karachi. NIO Tech Report.

INTERNATIONAL / NATIONAL OCEANOGRAPHIC RESEARCH CRUISES AND COASTAL ZONES FIELD PROGRAMS

In the following Geological and Geophysical/Oceanographic cruises participated as a Marine Geologist.

1. (2011) Geo-environmental studies of island and coastal mud volcanoes near Hingol river, Makran Coast
2. Survey of Offshore Island near Hingol Estuary, Makran.
3. (2002-2008) Geological Field mapping programs of M.Sc Classes in Cap Monze coastal area of Karachi.
4. (2005) Holocene Indus delta sediments studies.
5. (2003) Tasman spirit Oil Spill 13th August-2003 , Impacts on Beach Sediments. Karachi.
6. (March, 2002) Onboard S/V Behr Paima Geological and Geophysical research cruise off Pasni, Makran.
7. (2001) Study to Restore and Develop Bundal and Kiprianwala Islands. Pakistan Defense officers Housing Authority, Karachi.
8. (2000) Impact Assessment Study on Coastal Zones, Sea level Rise and Monsoon Variations. Pakistan National Communication
9. April 3-May 3, 1998. Onboard R/V "SONNE" Study of the Makran Accretionary wedge off Pakistan, Tectonic Evolution and fluid migration. Research cruise SO-130 organized by Federal institute for Geosciences and Natural Resources (BGR), Germany.
10. September-October, 1993. (Research cruise SO-90) Onboard R/V"SONNE", Geological studies of continental margin off Pakistan, PAKOMIN Program to investigate oxygen minimum zone and sediments depositing on the continental margin off Pakistan, organized by Federal institute of Geology and Natural resources (BGR), Germany.
11. October-November, 1992. "Leg D-2 Geological studies of Arabian Sea" Onboard R/V "TYRO", The Netherlands Indian Ocean Program (NIOP-92) organized by Marine Research foundation (SOZ), The Netherlands.
12. April 4-May 11, 1991. Onboard S/V"BEHR PAIMA" Pakistan. Oceanographic Survey of Gwadar and Giwani Areas, Arabian Sea. National Institute Of Oceanography, Pakistan.
13. March 21-29,1990. Onboard S/V "BEHR PAIMA" Pakistan. Geophysical and Geological Survey Offshore Indus Delta, Arabian Sea. National Institute Of Oceanography, Pakistan.

14. November-December, 1986. Onboard R/V "CHARLES DARWIN" UK. Geophysical and Geological Studies of Oman Basin, Owen Fracture Zone and Murray Ridge, Arabian Sea. Bullard Laboratories, Cambridge, U.K.
15. May 31-June 5, 1985. Onboard S/V "BEHR PAIMA" Pakistan. Geological and Sedimentological Survey of Murray Ridge, Arabian Sea. National Institute Of Oceanography, Pakistan
16. October-November, 1984. Onboard S/V "BEHR PAIMA" Pakistan. Hydrographic Survey of Gwadar, Ormara and Pasni Areas. Arabian Sea. National Institute Of Oceanography, Pakistan
17. April 8-May 10, 1984. Onboard S/V "BEHR PAIMA" Pakistan. Sedimentological Studies offshore Indus Delta. Arabian Sea. National Institute Of Oceanography, Pakistan
18. May, 1983. Onboard S/V "BEHR PAIMA" Pakistan. Physical oceanography cruises of Murray Ridge Area, Arabian Sea. National Institute Of Oceanography, Pakistan

COURSES/TRAINING

1. Remote sensing software (ER-MAPPER) training.. Engineering Services Limited, Karachi.
2. Short course on Concepts of Sequence Stratigraphy. Organized by Geomarine Division, National Science Foundation USA, & National Institute of Oceanography, Karachi, Pakistan.
3. Refresher course in secretariat procedure and practices.. Organized by Office management wing, National Institute of Public Administration (NIPA), Karachi.
4. Course on Isotope Geology. Organized by Grant Institute of Geology and Geophysics, University of Edinburgh. UK.
5. Course on X-Ray Diffractometry and X-Ray Fluorescence Techniques in Geology. Organized by Grant Institute of Geology and Geophysics, University of Edinburgh. UK.
6. Certificate Course in Computer Programming Fortran-77. Imperial College of Science and Technology, University of London. UK.
7. Advance Course in English Language. Cambridge School of Advance English, Cambridge. UK.
8. Advance Course in Oceanography. Organized by University of Hawaii, U.S.A. And National Institute of Oceanography, Pakistan.
9. Certificate Course in Composition, Properties and Functions of Drilling Fluids. Department of Geology, University of Karachi. Pakistan.

SEMINARS/CONFERENCES/WORKSHOPS

In the following international and national seminars/conferences and workshops presented scientific papers addressing important marine research problems of North Arabian Sea and Pakistan' EEZ.

1. Integrated Coastal Area Management Planning For Pakistan. 13-14 May, 2011. Organized by National Institute of Oceanography, Karachi. Pakistan

2. Marine Resource Potential and Environment of The Northern Arabian Sea. 30th April.2011. Organized by National Institute of Oceanography, Karachi. Pakistan
3. Technical Lecture Series # 9. 24th March, 2011. American Association of Petroleum Geologists (AAPG)-PAPG. Department of Geology, University of Karachi.
4. DelPHE Project Seminar (Biodiversity, Aquaculture and Fisheries) 20th April 2010. Centre of Excellence in Marine Biology, University of Karachi.
5. 35th All Pakistan Science Conference. December 20-23, 2008. Karachi Pakistan. University of Karachi Pakistan Society for Microbiology and Pakistan Association for the Advancement of Science.
6. Workshop on Ways and Means to overcome Energy Crisis in Pakistan. July 29-30, 2008. Karachi. Department of Geology, Federal Urdu University of Arts, Science and Technology, Karachi. Higher Education Commission (HEC), Islamabad and Society of Economic Geologists and Mineral Technologist.
7. Seminar on Coping with Coastal and Marine Challenges: Pan Asia Learning, July 14, 2008. Karachi. International Union for Conservation of Nature. Pakistan.
8. Seminar on Shrimp Fisheries of Pakistan. September 13' 2006. Centre of Excellence in Marine Biology, University of Karachi.
9. Workshop on Log frame and Proposal Writing, Karachi. May 10, 2006. Higher Education Commission of Pakistan and British Council Higher Education Links Programme.
10. Seminar on Seismic Risk of Karachi and its Mitigation ‘ Some new thoughts’. May 8, 2006. Geological Survey of Pakistan.
11. International Conference on Environment Horizons, Dec.19-21,2005. Department of Chemistry and International Center for Chemical Sciences. University of Karachi.
12. Seminar at the occasion of World Space week on Space science and The challenges of 21st Century. November 21st 2005. Institute of Space and Planetary Astrophysics.
13. Seminar on Earthquake Disaster, October 17, 2005. Department of International Relations. University of Karachi.
14. National Conference on Fossil Fuels and Mineral Resources of Sindh. 28th Feb-01March, 2005. Department of Geology , University of Sindh. Jamshoro.
15. Workshop on Marine Protected Area, 12th Feb.2005. Scottish Association For marine Sciences, U.K and Department of International Development , British Council Karachi
16. 5th International Conference on Asian Marine Geology IGCP-475 DeltaMap / APN Mega-deltas in Asia. 13-18 January 2004.Bangkok, Thailand.
17. Annual Technical Conference, “Knowledge Management Society of Petroleum Engineers and Pakistan Association of Petroleum Geoscientists” October 8-9, 2004. Islamabad.
18. 4th International Scientific Conference, October 6-9, 2004. University of Azad Jammu & Kashmir. Muzaffarabad.
19. JOI-USSAC Planning meeting of Scientific Drilling of the Indian Ocean Submarine Fans.23-27th July 2003. University of Colorado, Boulder. USA. Joint Oceanographic Institutions (JOI), Washington and Integrated Ocean Drilling Program (IODP).
20. Seminar on Prospects and Benefits of Deep Ocean Exploration and Mining for Pakistan. 26th June 2001. Organized by National Institute of Oceanography, Karachi. Pakistan.
21. Geologic and Climatic Evolution of Arabian sea region, 5-6th April 2001, London.U.K. Organized by The Royal Geological Society of London. U.K.

22. Symposium on "Untapped Fisheries Resources of Pakistan". 23-25 February 2001. Organized by Marine Reference Collection Center, University of Karachi and Pakistan Fisheries Society.
23. Seminar on Pak-US Cooperative Program in Oceanography 23-24th January 2001. Organized by National Institute of Oceanography, Karachi. Pakistan.
24. Fifth International conference on Trace Metals in the Aquatic Environment, 25-27th January 2001. Organized by National Institute of Oceanography, Karachi. Pakistan.
25. Workshop (14th March, 2000) on Marine Science Education in the 21st Century Present and Future Perspectives. Organized by the Scottish Association for Marine Science and British Council Karachi, Pakistan.
26. Statusseminar 1999 (10-12 March) Meeres Forschung mit FS SONNE. Freiberg, Germany. Organized by Federal Ministry for Research and Development (BMBF), Germany.
27. 3rd Pakistan Geological Congress 1997 (October 27-30). Peshawar University. Pakistan.
28. Workshop "Hydrography, biology, geochemistry and geology of the Arabian Sea and adjacent waters" 13-16 February 1995. Texel, The Netherlands. Organized by Netherlands Institute for Sea Research (NIOZ), Texel and Geosciences Foundation (GOA) of The Netherlands Organization for Scientific Research (NWO, The Hague).
29. International Workshop on integrated coastal zone management. 10-13 October 1994. Karachi, Pakistan. Organized by National Institute of oceanography. Sponsored by International Oceanographic Commission-UNESCO and UNDP.
30. PAK-US conference on Arabian Sea living resources and the environment. 20-24 June 1993. Karachi, Pakistan. Organized by American Institute of Marine Biology, Office of Naval Research U.S.A. and Marine Reference Collection and Resources Center, University of Karachi, Pakistan.
31. First South Asia Geological Congress, 23-27 February 1992. Islamabad, Pakistan. Organized by Ministry of Petroleum and Natural Resources, Pakistan.
32. Second IOC/WESTPAC Symposium, Scientific Contribution to the effective management of the Marine environments in the western Pacific. 2-6 December 1991. Penang, Malaysia. Organized by Ministry of Science and Environment, Malaysia and International Oceanographic Commission UNESCO.
33. National Seminar on study and Management in Coastal Zones in Pakistan. 23-26 September 1991. Karachi. Organized by Marine Reference Collection and Resources Center, University of Karachi and Pakistan National Commission for UNESCO.
34. International Symposium on the Indus " The Life line of Pakistan ". 2-7 March 1991. Islamabad. Organized by Pakistan council of Research for water resources.
35. PAK-US planning workshop on North Arabian Sea Environment and Ecologic Research Program. 7-10 January 1991. Karachi. Organized by Ocean Naval Research U.S.A. and National Institute of Oceanography, Pakistan.
36. First International Conference and Exhibition on Development of Ocean Resources for Economic Progress. 22-24 May 1991. Karachi. Organized by Ministry of Science and Technology Government of Pakistan and National Institute of Oceanography, Pakistan.
37. National Seminar on Universal Problems and Environmental Management. 20th March 1990. Organized by Scientific and Cultural Society of Pakistan.
38. Symposium on the Late Cenozoic Ice Age. 3-5th July 1989. Edinburgh, UK. Organized by Royal Society of Edinburgh. UK.

39. U.K. Oceanography '88'. 11-16 September 1988. Norwich, UK. Organized by Challenger Society, University of East Anglia. Norwich. UK.
40. International Congress of Geochemistry and Cosmochemistry. 29th Aug - 2nd Sept 1988. Paris, France. Organized by European Association For Geochemistry.
41. International meeting on the Geology and Tectonics of the Oman Region. 29-31 March 1988. Edinburgh, UK. Organized by Ministry of Petroleum and Natural Resources, Sultanate of Oman and Royal Society of Edinburgh, UK.
U.S. Pakistan Workshop in Marine Science, 8-12 November 1982. Karachi. Organized by National Science Foundation, U.S.A. and Ministry of Science and Technology. Government of Pakistan.
42. National Symposium on the role of Universities in Improving the Scientific Education and Research. 10-11 August 1982. Karachi. Organized by University of Karachi. Pakistan.

PUBLICATIONS

1. Naeem. Ahmed Syed; Arshad Ali; **Athar. AliKhan**; M.Ayub Yousufzai, **2006**. Diurnal Fluctuations of the Heat fluxes over the Northern Arabian Sea. Journal of Basic and Applied sciences. Vol.2 (1)
2. Arshad. Ali; **Athar. AliKhan**; **2006**. Arabian Sea Sub -Surface Salinity Minima (ASSM). Pakistan Journal Of Marine Sciences. Vol. 15 (1)
3. Naeem. Ahmed Syed; Arshad Ali; **Athar. AliKhan**; **2005**. The Study of formation of upper salinity maxima in the northern Arabian Sea and its distribution. Pakistan Journal Of Marine Sciences, Vol. 14 (1), 1-12.
4. Arshad Ali, **Athar AliKhan**, Naeem Ahmed Syed; **2005**. Mixed layer depths in northern Arabian sea. Pakistan Journal of Oceanography. Vol.1(1): 89-96.
5. Peter D. Clift, Ian H. Campbell, Xifan Zhang, Andrew Carter, Kip V. Hodges, **Athar Ali Khan** and Charlotte M. Allen. **2004** . Thermochronology of the Modern Indus River Bed load: New insight into the Controls on the Marine Stratigraphic Record. Tectonics, 23, No. 5
6. **Athar. AliKhan**; Nadeem.Ahmad, Khan; Tayyaba. Adil; Waqar. Majeed; **2004**. Litho- and Biofacies of the Laminated Hemipelagic Sediments from Upper Makran Slope, North Arabian Sea. Society of Petroleum Engineers and Pakistan Association of Petroleum Geoscientists Annual Technical Conference, "Knowledge Management" October 8-9, 2004. Islamabad.
7. Lee, J. I., P. D. Clift, G. D. Layne, J. Blum and **A. A. Khan**,. **2003** Sediment flux in the modern Indus River traced by the trace element composition of detrital amphibole grains, Sedimentary. Geology., 160, 243–257.
8. Arshad Ali, **Athar Ali Khan**, Naeem Ahmed Syed, **2003**. Spreading Of Persian Gulf Water In The Northern Arabian Sea. Pakistan Journal Of Marine Sciences, Vol.12 (2), 107-117.
9. Arshad Ali, **Athar Ali Khan**, **2003**. Variability Of Sea Surface Salinity In The Northern Arabian Sea. Pakistan Journal Of Marine Sciences, Vol. 12 (1), 29-38.
10. Peter D.Clift, Jae II Lee, Peter Hildebrand, Nobumichi Shimizu Graham D. Layne, Jerzy Blusztajn, Joel D.Blum, Eduardo Garzanti, **Athar Ali Khan**, **2002**. Nd and Pb Isotope variability in the Indus River System: implications for sediment provenance and crustal

- heterogeneity in the Western Himalaya. *Earth and Planetary Science Letters* 200 (2002) 91-106.
11. U.VonRad, **Athar. AliKhan**, W.H.Berger, D.Rammlmair, U.Treppke, **2002**. Varves, Turbidites and Cycles in Upper Holocene Sediments (Makran Slope, northern Arabian Sea). Clift, P. D; Kroon, D; Gaedick, C; Craig, J. (eds) 2002. The Tectonic and Climatic Evaluation of the Arabian Sea Region. Geological Society, London, Special Publication, 195, 387-406.
 12. Klaus-Peter. Burgath, Ulrich. VonRad, Willem, Vander Linden, Martin. Block, **Athar. Ali Khan**, Hans. Albert Roeser, wolfgang. Weiss, **2002**. Basalt and Peridotite recovered from Murray Ridge: Are they of supra-subduction origin ? Clift, P. D; Kroon, D; Gaedick, C; Craig, J. (eds) 2002. The Tectonic and Climatic Evaluation of the Arabian Sea Region. Geological Society, London, Special Publication, 195, 117-135.
 13. Andreas. Luckge; H.Doose, **Athar.Ali Khan**, H.Schultz, U.vonRad, **2001**. Monsoonal variability in the northeastern Arabian Sea during the past 5,000 years: Geochemical evidence from laminated sediments. *Plaeogeography, Paleoclimatology, Palaeocology* 167, 273-286.
 14. VonRad.U, **Athar.Ali Khan**, W.H.Berger, D.Rammlmair and U.Treppke, **2001**. Varve, Turbidites and cycles in upper Holocene sediments Makran slope, North Arabian Sea. Clift, P. et al. (eds.), *Geologic and Climatic Evolution of the Arabian Sea Region*, Geol. Soc. (London), Special Publication. (2001).
 15. Burgath , K.P, U. von Rad, W.Van der Linden, M.Block, **A. Ali Khan**, H.A. Roeser and W. Weiss, **2001**. Basalt and Peridotite Recovered from Murray Ridge: Are they of Supra-Subduction Origin? Clift, P. et al. (eds.), *Geologic and Climatic Evolution of the Arabian Sea Region*, Geol. Soc. (London), Special Publication (2001).
 16. Jae II Lee, Peter Clift, Jerzy Blusztajn, Graham Layne, Joel Blum, **Athar Ali Khan; 2001**. Modest Sediment Flux From Nanga Parbat in to the Indus River traced by Amphiboles and Nd Isotopes. Submitted to *Geology*.
 17. **Athar. A. Khan**; C.H.Weijdan; Reihart, Gert Jan; Henrich, V; **2001**. Organic matter variations in northwestern Arabian Sea sediments. *Pakistan Journal Of Marine Sciences* Vol, 10. (1),1-14.
 18. **Athar. Ali .Khan**, Gulraiz Hamid Rizvi, Syed Imran Hasnay. **2000**. Mineralogical variations and physical properties of deep-sea sediments, Northwestern Arabian Sea. *Pakistan Journal Of Marine Sciences*, Vol.9 (1&2),17-31.
 19. U.Von Rad, Berner. U, G.Delisle, H.Doose, N.Fechner, P.Linke, A.Luckge, H.A.Rosser, R.Schmaljohann, M.Wiedcke, M.Block, **A.Alikhan**, V.Damm, J.Fritsh, A.Inam, A.H.Cheema, M.Tahir, A.R.Tabrez, M.Ashraf, M.Prevaz, B.Schreckenberger, M.Drews, J.Erbacher, B.Harzim, N.V.Mirbach, J.Poggenburg, G.Scheeder, S.Walter, **2000**. Gas and Fluid venting at the Makran Accretionary wedge off Pakistan: Initial Reports. *Geomarine Letters* 20, 10-19.
 20. **Athar. A. Khan; 1999**. Accumulation of Organic carbon in northwestern Arabian Sea. *Pakistan Journal of Hydrocarbon Research*. Vol.11, p.51-58.
 21. **Athar. A. Khan; 1999**. Offshore Geology of Pakistan and non-living resource prospects. *Pakistan Journal of Marine Sciences*. Vol.8 (1), 81-97.
 22. **Athar. A. Khan; 1997**. Quartz and dolomite content variations and climate changes in Late Pleistocene sediments of the North Arabian Sea. *Geological Bulletin, University of Peshawar*.Vol.30, 1-11.

23. U.Von Rod, H.Schultz, **Athar.AliKhan**, A.Ansari, M.Berner, U.Cepeck, G.Cowie, P.Dietrich, H.Erlenkeuser, M.Geyh, M.Jennerjahn, A.Lukge, V.Marching, V.Riech, V.Rosch, H.Schafer, P.Schulte, F.Sirocko, M.Tahir, M.Weiss, **1995**. Sampling the oxygen minimum zone off Pakistan: glacial-interglacial variations of Anoxia and productivity. *Marine Geology*.125, 7-19.
24. **Athar. A. Khan; 1995**. Monsoon changes and palaeoproductivity in Northern Arabian Sea sediments. In: Thompson, M.F and Tirmizi, N.M; (eds) *Arabian Sea living resources and the environment*. Vanguard Books (Pvt) Ltd, Lahore. Pakistan.
25. **Athar. A. Khan; L.I.Kazi; A.Ali; M.Ali; 1993**. Global Warming and its possible effects along Sonmiani and Gadani, Baluchistan coast. In proceedings of National conference on Problems and Resources of Makran Coast and Plan of Action for its Development. 28-30 September, 1991. Quetta, Pakistan. Organized by Pakistan Council for Scientific and Technology.
26. **Athar. A. Khan; G.M. Memon; M.Danish; A.Inam; 1993**. Distribution of surface sediments off Indus Delta on the continental shelf of Pakistan. *Pakistan Journal of Marine Science*. Vol.(2),1
27. **Athar. A. Khan; L.I.Kazi; A.Ali; M.Ali; 1993**. Global Warming and its impacts along the Makran Coast. Pakistan. In Proceedings of National Seminar on study and Management in Coastal Zones in Pakistan. 23-26 September, 1991. Karachi. Organized by Marine Reference Collection Center, University of Karachi and Pakistan National Commission for UNESCO.
28. Asif Inam; **A. A. Khan; 1993**. Mud Volcanoes along Pakistan Coast. In proceedings of National conference on Problems and Resources of Makran Coast and Plan of Action for its Development. 28-30 September, 1991. Quetta, Pakistan. Organized by Pakistan Council for Scientific and Technology.
29. **Athar. A. Khan; N.B.Price; 1991**. Climatic changes during last 250.000 years BP and its influence on the geochemical fluxes in sediments of northern Arabian Sea. In proceedings of Second IOC/WESTPAC Symposium, Scientific Contribution to the effective management of the Marine environments in the western Pacific. UNESCO, International Oceanographic Commission.
30. **Athar. A. Khan; 1990**. Mineral wealth in the submerged hills and basins of Pakistan EEZ. Sahil, Special Volume. NIO Bulletin.

PUBLISHED RESEARCH PAPERS ABSTRACTS IN PROCEEDINGS OF WORKSHOPS/SEMINARS/SYMPOSIUMS

1. **Athar. Ali Khan, 2009**. Paleoclimatic variability record from varved sediments off Pakistan continental margin. 35th All Pakistan Science Conference. December 20-23, 2008. Karachi Pakistan. University of Karachi Pakistan Society for Microbiology and Pakistan Association for the Advancement of Science
2. **Athar. Ali Khan, 2009**. Offshore mineral resource prospects of Pakistanis Economic Exclusive zone. 35th All Pakistan Science Conference. December 20-23, 2008. Karachi Pakistan. University of Karachi Pakistan Society for Microbiology and Pakistan Association for the Advancement of Science
3. **Athar. Ali Khan, Tayyaba. Adil, S.A. Akbar Adil, Mujeeb.Ahmad, Nadeem. Ahmad Khan, S. Amir A. Shah, 2005**, Gas-Hydrates: Potential Future Energy Resources of

- Pakistan. National Conference on Fossil Fuels and Mineral Resources of Sindh. 28th Feb-01 March, 2005. Department of Geology, University of Sindh. Jamshoro
4. Peter D. Clift, Xifan Zhang, Kip V. Hodges, **Athar Ali Khan**; 2004. The Marine Record of Continental Erosion and Cenozoic Climate change in the Asian Marginal Seas. 5th International Conference on Asian Marine Geology IGCP-475 DeltaMap / APN Mega-deltas in Asia. 13-18 January 2004. Bangkok, Thailand
 5. **Athar Ali Khan**, Ulrich. von Rad, Andreas. Luckge, 2004. High Resolution Paleoclimate Variability documented in Upper Holocene Varved Sediments off Pakistan. 5th International Conference on Asian Marine Geology IGCP-475 DeltaMap / APN Mega-deltas in Asia. 13-18 January 2004. Bangkok, Thailand
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